

1(a)(i)	<i>independent variable</i> the side / surface of the leaf ; <i>dependent variable</i> distance / how far the water moves (down) / the decrease (in volume) of the water in the graduated tube AW ;	2
1(a)(ii)	<i>any six from</i> 1 ref. to cutting plant underwater / airtight seal on cup ; 2 use upper and lower surfaces ; 3 suck water into graduated tube and then close / attach clip ; 4 record the distance moved / volume change at start and end of a time ; <i>standardised variables (mp 5 to mp 9) max 2</i> 5 method of keeping same light (intensity) ; 6 method of keeping same temperature ; 7 same humidity ; 8 method of keeping same air flow ; 9 same leaf / type of leaf / species of leaf / use the same plant ; 10 idea of re-setting water level ; 11 to obtain at least 3 results for each leaf surface to obtain a mean / identify anomalies ; 12 low risk / medium risk experiment / stated risk and precaution ;	6

Question	Answer	Marks
1(b)(i)	<i>ref. to</i> measure radius of the area of leaf under the cup ; <i>ref. to</i> (using) πr^2 ; or <i>ref. to</i> using a (transparent) grid on the area under the cup ; count squares ;	2
1(b)(ii)	<i>any three from</i> 1 divide water loss by area or by time ; 2 divide result (from mp 1) by the variable not used in mp 1 ; 3 calculate the volume of water loss ; 4 <i>ref. to</i> comparing (measurements) from both sides ; 5 <i>ref. to</i> statistical test / <i>t</i> -test (to see if difference is significant) ;	3
1(b)(iii)	<i>correct axis labels:</i> x-axis: minimum: leaf surface, y-axis water (vapour) loss / movement ; y-axis: correct volume or distance unit and time unit or area unit on y-axis ; bar(s) are identified and upper leaf surface bar is smaller ;	3

Question	Answer	Marks																																			
1(c)(i)	<i>any two from</i> <table><tr><th rowspan="2">experimental condition</th><th colspan="5">distance moved by water column / cm per unit time</th></tr><tr><th>trial 1</th><th>trial 2</th><th>trial 3</th><th>trial 4</th><th>trial 5</th></tr><tr><td>high light intensity</td><td></td><td></td><td></td><td>6.2</td><td></td></tr><tr><td>no light</td><td></td><td></td><td>1.5</td><td></td><td></td></tr><tr><td>high temperature</td><td>4.4</td><td></td><td></td><td>4.8</td><td></td></tr><tr><td>strong air current</td><td></td><td>2.4</td><td></td><td></td><td></td></tr></table> ;;	experimental condition	distance moved by water column / cm per unit time					trial 1	trial 2	trial 3	trial 4	trial 5	high light intensity				6.2		no light			1.5			high temperature	4.4			4.8		strong air current		2.4				2
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1(c)(ii)	<i>any one from</i> high light intensity causes aperture to be wider ; ora high temperature causes aperture to be wider than strong air currents / no light ; ora	1																																			